

GDYNIA OKSYWIE, Poland

WMO: 121420

Lat: 54.5857N

Lon: 18.5192E

Elev: 131

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.8	(c) 16.4	(d) 6.5	(e) 7.7	(f) 13.9	(g) 10.4	(h) 9.4	(i) 17.9	(j) 36.1	(k) 39.1	(l) 31.7	(m) 39.2	(n) 7.6	(o) 150	(p) 0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	79.1	68.0	76.2	66.7	73.6	65.5	70.3	76.4	68.4	74.0	66.7	71.8	9.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	103.3	74.1	66.3	97.2	71.7	64.5	91.5	70.1	34.3	76.6	32.7	74.1	31.4	71.7	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.4	22.5	19.4	DB	5.8	87.1	7.2	3.2	0.7	89.4	-3.6	91.2	-7.6	93.0	-12.8	95.3	
			WB	5.8	73.5	6.9	2.2	0.8	75.0	-3.2	76.3	-7.1	77.5	-12.1	79.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	30.6	32.2	37.5	45.0	53.3	60.1	64.6	64.5	57.9	49.0	41.3	35.5
	DBStd	13.39	8.35	7.40	6.98	6.40	6.27	5.18	4.42	4.39	4.50	5.88	5.38	7.55
	HDD50	2509	601	499	389	177	33	1	0	0	2	92	264	451
	HDD65	6431	1065	919	851	601	367	165	62	61	216	497	712	915
	CDD50	1672	0	0	3	26	135	302	453	451	240	60	2	0
	CDD65	123	0	0	0	0	5	17	50	47	4	0	0	0
	CDH74	528	0	0	0	1	28	98	209	168	24	0	0	0
	CDH80	74	0	0	0	0	3	17	30	21	3	0	0	0
Wind	WSAvg	9.2	11.1	10.4	10.3	8.7	8.0	8.5	7.9	7.7	8.2	9.1	9.7	10.6
Precipitation	PrecAvg	22.9	1.4	1.1	1.2	1.2	1.9	2.3	3.1	2.6	2.4	2.1	1.8	1.7
	PrecMax	28.2	3.2	2.1	2.1	2.9	3.7	5.1	6.9	5.7	5.8	4.8	4.1	2.7
	PrecMin	17.0	0.2	0.1	0.2	0.1	0.6	0.3	0.5	0.2	0.5	0.5	0.3	0.7
	PrecStd	3.0	0.7	0.5	0.5	0.7	0.9	1.1	1.4	1.5	1.3	1.1	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	50.2	60.2	70.1	77.7	82.8	83.7	82.7	78.2	67.2	55.3	52.7
		MCWB	47.9	46.1	49.9	55.2	64.3	68.5	69.8	69.1	64.3	59.2	50.5	49.1
	2%	DB	46.6	46.9	53.8	63.8	72.1	76.5	78.9	78.1	71.2	63.2	52.6	49.2
		MCWB	44.0	42.9	46.7	52.8	62.4	65.3	69.1	67.7	62.3	57.4	49.7	46.6
	5%	DB	43.8	44.0	50.1	58.9	68.0	72.5	75.9	75.2	68.2	60.2	50.6	46.8
		MCWB	41.6	40.9	44.4	50.4	60.3	63.9	67.5	66.5	61.1	55.3	48.3	44.3
	10%	DB	41.6	41.8	47.0	55.3	64.0	69.3	73.3	72.8	65.8	57.5	48.7	44.5
		MCWB	39.6	39.1	42.7	48.5	57.9	61.3	66.1	65.5	59.8	53.9	46.6	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	47.8	51.4	57.4	66.3	71.1	73.1	72.1	67.3	60.4	52.0	49.8
		MCDB	50.6	49.0	56.8	66.3	74.3	80.0	80.5	78.5	74.0	64.7	53.7	52.1
	2%	WB	44.3	43.2	48.0	53.8	63.5	67.2	70.8	69.8	64.0	58.1	50.1	47.1
		MCDB	46.1	45.7	52.6	62.8	70.6	74.2	76.8	75.4	68.7	62.2	52.1	49.1
	5%	WB	41.8	41.2	45.1	51.2	61.0	64.8	68.9	68.0	62.2	55.9	48.5	44.5
		MCDB	43.6	43.9	48.9	57.6	67.0	70.9	74.2	73.5	66.8	59.5	50.4	46.2
	10%	WB	39.6	39.1	43.0	49.1	58.2	62.6	67.1	66.3	60.6	54.1	46.8	42.5
		MCDB	41.5	41.5	46.5	54.3	63.4	67.8	71.8	71.2	65.1	57.1	48.4	44.4
Mean Daily Temperature Range	5% DB	MDBR	7.1	8.3	10.8	14.3	15.0	14.8	13.4	14.0	12.6	10.7	7.5	7.1
		MCDBR	7.7	10.4	16.5	20.4	20.4	19.9	17.6	17.5	16.5	14.6	9.3	8.2
		MCWBR	6.9	8.0	10.7	12.2	11.8	11.3	9.8	9.7	9.8	9.7	7.3	7.4
	5% WB	MCDBR	7.6	10.2	14.5	19.0	18.6	17.8	15.8	15.7	15.1	13.4	9.0	7.9
		MCWBR	7.1	8.2	10.3	12.0	11.4	11.2	9.8	9.8	9.9	9.5	7.4	7.5
Clear-Sky Solar Irradiance	taub		0.311	0.339	0.365	0.408	0.399	0.395	0.425	0.418	0.378	0.360	0.344	0.310
	taud		2.288	2.293	2.289	2.226	2.323	2.366	2.298	2.321	2.410	2.429	2.327	2.287
	Ebn at Noon		198	233	255	258	269	271	259	253	249	225	183	171
	Edh at Noon		18	26	32	39	37	36	38	35	29	23	18	15
All-Sky Solar Radiation	RadAvg		145	357	757	1336	1741	1825	1636	1397	985	495	187	106
	RadStd		19	55	85	161	168	149	180	136	107	77	24	14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (24 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-234	+121	N/A